# AquaWatch AI

ENVIRONMENTAL WATER QUALITY MONITORING NETWORK - MAHARASHTRA

Official Laboratory Water Testing & Certified Machine Learning WQI Assessment

**LABORATORY REPORT**
**ID: REP-20261104-D12D35**
Status: VERIFIED AUDIT

## 1. SAMPLE & GEOGRAPHICAL INFORMATION

**Sample ID:** SMP-20261104-K2JY
**Water Body:** Panchganga River - Kolhapur (River)
**Location:** Shiroli Bridge, Kolhapur, Kolhapur, Maharashtra
**Coordinates:** 16.7050 N, 74.2433 E
**Sample Type:** Surface Water | Sampled: 04 Nov 2026, 11:55

## 2. LABORATORY & TESTING INFORMATION

**Laboratory:** AquaWatch Central Water Testing La
**Technician:** Dr. Sunita Patil (Lab Authority)
**Testing Date:** 04 Nov 2026, 00:00
**Test Method:** Standard Methods (APHA / IS 3025)
**Remarks:** Sample collected under certified env

**SAFETY STATUS**
**HARMFUL**

**PREDICTED WATER QUALITY INDEX (WQI)**
**129.17** / 100 Scale

**GRADE & MODEL SPECIFICATION:**
**Grade: Severely Polluted (Grade E - Unfit for Drinking)**
ML Algorithm: Decision Tree (12 Features)

## 3. COMPLETE 14-PARAMETER ANALYTICAL MEASUREMENT RESULTS

| No. | Water Quality Parameter | Unit | Regulatory Baseline (CPCB/IS 10500) | Measured | Evaluation |
|---|---|---|---|---|---|
| 1 | **pH** | pH | 6.5 - 8.5 (CPCB Class C / IS 10500) | **5.20** | FAIL |
| 2 | **Dissolved Oxygen (DO)** | mg/L | >= 4.0 (CPCB Minimum) | **3.00** | FAIL |
| 3 | **Biochemical Oxygen Demand (BOD)** | mg/L | <= 3.0 (CPCB Limit) | **8.00** | FAIL |
| 4 | **Chemical Oxygen Demand (COD)** | mg/L | <= 250.0 (General Effluent / Surface) | **45.00** | PASS |
| 5 | **Turbidity** | NTU | <= 5.0 (IS 10500 Permissible) | **25.00** | FAIL |
| 6 | **Total Dissolved Solids (TDS)** | mg/L | <= 500.0 (IS 10500 Acceptable) | **900.00** | FAIL |
| 7 | **Electrical Conductivity (EC)** | uS/cm | <= 1500.0 (IS 10500 / Irrigation) | **1100.00** | PASS |
| 8 | **Temperature** | deg C | Ambient (15.0 - 35.0 deg C) | **26.50** | PASS |
| 9 | **Nitrate** | mg/L | <= 45.0 (IS 10500 Permissible) | **6.50** | PASS |
| 10 | **Phosphate** | mg/L | <= 0.10 (Oligotrophic Limit) | **0.25** | FAIL |
| 11 | **Chloride** | mg/L | <= 250.0 (IS 10500 Acceptable) | **180.00** | PASS |
| 12 | **Total Coliform** | MPN/100 mL | <= 5000 (CPCB Class C Max) | **3500.00** | PASS |
| 13 | **Fecal Coliform** | MPN/100 mL | <= 500 (CPCB Class C Desirable) | **1200.00** | FAIL |
| 14 | **Sulphate** | mg/L | <= 200.0 (IS 10500 Acceptable) | **150.00** | PASS |

## 4. OBSERVATIONS & REMEDIATION RECOMMENDATIONS

- Parameter threshold exceedances detected; water is currently NON-COMPLIANT for direct or untreated use.
- Mandatory coagulation, flocculation, advanced multi-barrier filtration, and robust UV/chlorination required.
- Trace pollution source upstream and re-sample within 48 hours to confirm contaminant levels.

**Certified Laboratory Technician:**
**Dr. Sunita Patil (Lab Authority)**
Digital Verification Hash: OK-VERIFIED-AUTH

**AquaWatch Quality Assurance Board:**
**State Water Quality Monitoring Cell**
Generated: 04 Nov 2026 17:29:46 UTC

NOTICE & REGULATORY DISCLAIMER: This test certificate is issued based on physical-chemical and biological laboratory analysis evaluated agai
nst CPCB Class C criteria and BIS IS 10500:2012 standards. WQI is a scientific indicator computed via machine learning and does not replace medical safety certifications. AquaWatch AI (c) 2026. All rights reser